12.5 Volumes of Pyramids and Cones

Name _____ Period ____ Date _____

Geometry 12.5 · Student sheet

I can…

- find the volume of a pyramid.
- find the volume of a cone.
- find the volume of a composite solid.

Vocabulary · fill in the missing word

Height of a pyramid or cone — the _____ distance from the vertex to the plane of the base

Composite solid — a solid made up of two or more _____

Part 1 · Discovery: how many cones fill the cylinder?

Fill the cone (or pyramid) and pour it into the cylinder (or prism) with the same base and height. Count the pours.

Solids used	Pours to fill	Fraction of the cylinder or prism

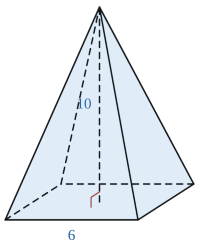
Volume of a cone or pyramid = _____ × volume of the matching cylinder or prism.

Part 2 · Notes and examples (fill in each note, then work the example)

Volume of a pyramid: V = _____ Bh

Example 1

Find the volume of the square pyramid.



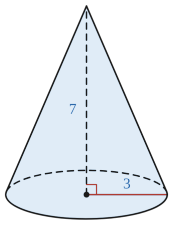
Example 2

A pyramid with a 6-by-4 rectangular base has a volume of 96 cubic feet. Find its height.

Volume of a cone: V = _____

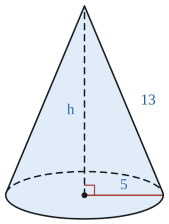
Example 3

Find the volume of the cone. Round to the nearest hundredth.



Example 4

A cone has a radius of 5 and a slant height of 13. Find its volume. Round to the nearest hundredth.



Composite solids: find the volume of each part, then _____ (or subtract a part that is removed).

Example 5

A grain silo is a cylinder with a radius of 3 meters and a height of 10 meters, topped by a cone with a height of 4 meters. Find its volume. Round to the nearest hundredth.

Part 3 · Partner practice

Partner 1

A pyramid has a square base with sides of 9 and a height of 8. Find its volume.

Partner 2

A cone has a diameter of 8 and a height of 9. Find its volume. Round to the nearest hundredth.